JOURNAL OF THERMOPHYSICS AND HEAT TRANSFER
Vol. 33, No. 4, October–December 2019

Check for updates

Free Convection from a Heated Vertical Cone in Generalized Newtonian Fluids

P. Mishra*
Indian Institute of Technology, Kanpur 208 016, Uttar Pradesh, India
and
N. Nirmalkar† and R. P. Chhabra‡
Indian Institute of Technology, Ropar 140 001, Punjab, India

DOI: 10.2514/1.T5614

In many situations, heat transfer occurs primarily by free convection, such as that in confined and restricted spaces and/or where it is not possible to induce fluid motion by using a pump or impeller. In this work, natural convection from an isothermal cone in power law and Bingham plastic fluids has been investigated numerically by solving the coupled momentum and energy equations over wide ranges of conditions such as the Grashof number ($10 - 10^4$), the Prandtl number (5–100), the Bingham number (1–50), and the power-law index (0.4–1). The structure of the flow and temperature fields is visualized in terms of streamlines and isotherm contours, yielded and unyielded regions, and the variation of the local Nusselt number along the surface of the cone. The location and shape of the yielded and unyielded regions are influenced by the values of the Grashof and Bingham numbers. The size of the unyielded regions progressively decreases with the increasing Bingham number, whereas the Grashof number promotes the growth of fluidlike regions. Shear-thinning fluid behavior promotes heat transfer with reference to that in Newtonian fluids. On the other hand, the Bingham yield stress has an adverse influence on the overall heat transfer. The present values of the average Nusselt number are correlated in terms of the pertinent dimensionless numbers to enable its prediction in a new application. Finally, a criterion for the onset of convection in Bingham fluid is also reported in terms of $Bn_{\max} \approx 0.15\sqrt{Gr_{\mathrm{BP}}}$; above which, conduction is the main mode of transport. Thus, the relative magnitudes of the buoyancy and viscous (including the contribution of yield stress) forces determine the degree of deterioration in the rate of heat transfer with reference to that in Newtonian fluids otherwise under identical conditions. This information is valuable: not only in designing heat transfer equipment, but also in the processing of temperature-sensitive materials.

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

Nomenclature

a, a^*	=	fitted coefficients in Eqs. (22) and (24)
Bn	=	Bingham number
b, b^*	=	fitted coefficients in Eqs. (22) and (24)
C_D	=	drag coefficient
c_p	=	specific heat of fluid, $\mathrm{J \cdot kg^{-1} \cdot K^{-1}}$
d	=	diameter of cone, m
F_D	=	drag force, N
Gr	=	Grashof number
Gr_L	=	Grashof number based on the slant length of the cone
Gr_L^*	=	$Gr_L \times \cos\phi$
g	=	acceleration due to gravity, $\mathrm{m \cdot s^{-2}}$
h	=	local heat transfer coefficient, $\mathrm{W \cdot m^{-1} \cdot K^{-1}}$
k	=	thermal conductivity of fluid, $\mathrm{W \cdot m^{-2} \cdot K^{-1}}$
M	=	regularization parameter
m	=	consistency index in power-law fluid, $\mathrm{Pa \cdot s^n}$
Nu_{avg}	=	average Nusselt number
Nu_L	=	local Nusselt number
Nu_{avg}^*	=	average Nusselt number based on slant length of the cone
n	=	power-law index
Pr	=	Prandtl number
p	=	pressure
R_∞	=	radius of spherical computational domain
r	=	radial coordinate, m
T	=	temperature of the fluid, K
T_w	=	temperature on the surface of the cone, K
T_0	=	temperature of the fluid far away from the cone, K
$\boldsymbol{U}$	=	velocity vector
U_r	=	r-component of the velocity
U_z	=	z-component of the velocity
β	=	coefficient of volumetric expansion, $\mathrm{K^{-1}}$
$\dot{\gamma}_{ij}$	=	rate of strain tensor
ϵ	=	regularization parameter in Bercovier and Engelman model
Θ	=	dimensionless temperature; $(T - T_0)/(T_w - T_0)$
$\Theta'(0)$	=	dimensionless temperature gradient on the surface of the cone
λ	=	coefficient in Eq. (8)
μ_B	=	Bingham plastic viscosity, $\mathrm{Pa \cdot s}$
μ_Y	=	yielding viscosity, $\mathrm{Pa \cdot s}$
ρ	=	density of the fluid, $\mathrm{kg \cdot m^{-3}}$
ρ_0	=	density of fluid at the reference temperature T_0, $\mathrm{kg \cdot m^{-3}}$
τ_{ij}	=	extra stress tensor, Pa
τ_0	=	Bingham yield stress, Pa
ϕ	=	half-angle of the cone, deg
II	=	second invariant of a tensor

Subscripts

BP	=	Bingham plastic
ij	=	r–z coordinates
PL	=	power-law fluid
w	=	cone surface condition

Received 29 August 2018; revision received 10 February 2019; accepted for publication 10 February 2019; published online 12 April 2019. Copyright © 2019 by the American Institute of Aeronautics and Astronautics, Inc. All rights reserved. All requests for copying and permission to reprint should be submitted to CCC at www.copyright.com; employ the eISSN 1533-6808 to initiate your request. See also AIAA Rights and Permissions www.aiaa.org/randp.

*Graduate Student, Department of Chemical Engineering.
†Assistant Professor, Department of Chemical Engineering.
‡Professor, Chemical Engineering.

I. Introduction

HEAT transfer by free convection from heated objects of a range of shapes in Newtonian fluids has been studied extensively over the past hundred years or so. The resulting voluminous body of

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

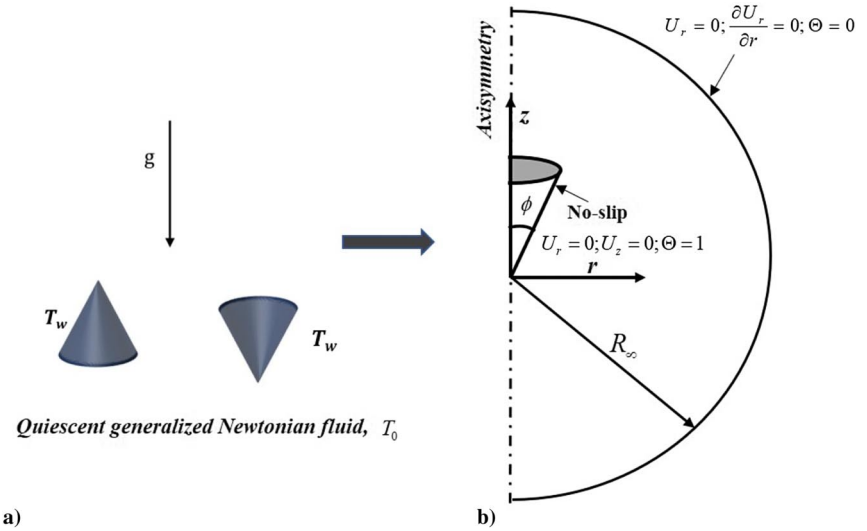
knowledge has been reviewed extensively by, among others, Martynenko and Khramstov [1], Oosthuizen and Kalender [2], etc. The bulk of the available literature deals with spheres and spheroids; hemispheres; disks and plates; two-dimensional cylinders of various cross sections, including circular, square, and triangular cross sections; and many other two-dimensional and three-dimensional shapes. All in all, based on a combination of the approximate analytical (boundary-layer equations), limited numerical and experimental results, reliable expressions are now available for estimating the value of Nusselt number for given values of Grashof and Prandtl numbers over the range of conditions of practical interest. Furthermore, limited results are also available on the delineation of different flow regimes (laminar and turbulent), on the role of confinement and temperature-dependent properties, etc. This body of knowledge has been reviewed in Ref. [1] and by others [3–8]. A cursory inspection of the available literature reveals that the case of a cone has received very little attention, even in Newtonian fluids. This geometry not only belongs to the general class of axisymmetric flows but also has relevance in the context of a cone penetrometer to determine the properties of soft solids and gels encountered in food, pharmaceuticals, personal care products, and construction materials. Indeed, this configuration is employed to delineate the nonisothermal effects (like effect of temperature on the firmness of butter [9] and on the strength of soils [10] due to the seasonal variation in temperature). Aside from these applications, a systematic study of free convection from such idealized shapes, shown schematically in Fig. 1a, also adds to our knowledge base on heat transfer in non-Newtonian fluids in the free and forced convection regimes [11–15].

On the other hand, it is readily acknowledged that most fluids of high molecular weight (polymers, resins, and gums), a multiphase nature (slurries, suspensions, foams, and emulsions), and containing large surfactant molecules (wormlike micelles), etc., exhibit a range of non-Newtonian characteristics, with the most common being shear-thinning viscosity, yield-stress, and viscoelasticity [16–18]. Such non-Newtonian fluid characteristics significantly influence the efficacy of heating, cooling, homogenization, atomization, spraying, etc., of non-Newtonian fluids in a broad spectrum of industrial settings. Typical examples include food, pharmaceuticals, personal care product sectors [19,20], polymer processing, advanced aviation fuels, drilling muds, electronic cooling and packaging operations, design of atomization and combustion nozzles using smart fuels, etc. [21–23]. Consequently, over the years, a significant research effort has been expended in investigating the role of shear thinning and yield stress on the processing of such materials in scores of geometries [16–18,21–23] and in numerous industrial applications related to enhanced oil recovery [24,25], convective momentum transport in porous media or in tubes with porous boundaries [26,27], etc. However, it is perhaps fair to say that much of the pertinent literature relates to the fluid mechanical aspects followed by forced heat transfer processes and, indeed, only limited studies are available on free convection, even for the highly idealized model configurations such as a sphere or a spheroid [28–30]. Suffice it to say here that, owing to their intrinsically viscous nature and/or yield stress, heat transfer in such fluids tends to be far more problematic than that in Newtonian fluids like air and water. Based on the currently available results for spheres, spheroids, and hemispheres [28–33], suffice it to add here that shear-thinning viscosity facilitates heat transfer by up to 50% over and above that in Newtonian fluids at the same values of Grashof and Prandtl numbers. On the other hand, the fluid yield stress has a deleterious effect on the overall free convection heat transfer from spheres, cylinders, spheroids, etc. This work is thus concerned with the laminar free convection from a 60 deg cone with its apex pointing, both in the upward and downward directions, in Newtonian, power-law, and Bingham plastic fluids to span a broad range of fluid behaviors.

Before a detailed presentation and discussion of the new numerical results gleaned in this work, it is useful to briefly review the corresponding state of the art in this area. Early attempts at studying natural convection heat transfer from a heated cone were based on the similarity solution of the boundary-layer equations in the limit of high Grashof and Prandtl numbers, thereby assuming the boundary layers to be thin. This, in turn, leads to the assumption of negligible curvature effects. This approximation is also approached by considering a slender cone that resembles a vertical plane surface. For instance, Hering and Grosh [34] reported approximate results on the Nusselt number for a vertical right circular cone based on the numerical solution of the transformed boundary-layer equations for a Prandtl number of 0.7 for both cases of constant and linear temperature distributions along the slanted surface of the cone. Their results for the average Nusselt number for an isothermal vertical cone (apex pointing upward) are given as follows:

$$Nu^*_{\text{avg}} = 0.51554 Gr_L^{*1/4} \tag{1}$$

On the other hand, Kuiken [35] analyzed the effect of curvature on the overall heat transfer by employing the power series expansion of the boundary-layer equations. He reported that the curvature effects are significant, even for a slender cone; and this simplification leads to the underestimation of the average Nusselt number. Similarly, the effect of curvature was also investigated by Oosthuizen and Wu [36] using a finite difference solution of the boundary-layer equations. Subsequently, Stewart [37] solved the three-dimensional boundary-layer equations for free convection transfer using a combination of variables in the limit of large Grashof and Prandtl numbers. For a vertical isothermal cone with its base insulated, the expression for the average Nusselt number is given as follows:



a) b)

Fig. 1 Schematic diagrams: a) flow domain, and b) computational domain.

$$Nu^{*}_{\text{avg}} = -\Theta'(0)(Gr_L Pr)^{1/4}\frac{2.43\sin\phi}{(\cos\phi)^{1/4}} \tag{2}$$

Alamgir [38] used an approximate integral method to evaluate the average Nusselt number from a slender cone with power-law temperature variation along the slanted surface of the cone. For the limiting case of an isothermal cone, the average Nusselt number is given as follows:

$$Nu^{*}_{\text{avg}} = \frac{16}{7}\frac{Gr_L^{*1/4}}{A} \tag{3}$$

where the values of A for a range of conditions can be found in the original publication [38]. Na and Chiou [39], Choiu and Na [40], and Na and Chiou [41] obtained an approximate solution based on a hybrid approach employing the combination of quasi-linearization and finite difference methods applied to the boundary-layer equations for frustums of the cone for the boundary conditions of uniform and variable temperatures along the surface of the cone. Hasan and Mujumdar [42] reported a numerical solution of the coupled nonlinear boundary-layer equations for natural convection and mass transfer from a vertical cone via the Newton–Raphson shooting method. Detailed results on the local wall shear stress and the local Nusselt/Sherwood number for a range of values of Schmidt numbers are reported. Pop and Watanabe [43] used the finite difference method to solve boundary-layer equations from a vertical cone in the presence of uniform injection/suction of the fluid with the constant heat flux condition prescribed on the surface of the cone. Furthermore, the transient laminar free convection from a vertical cone with uniform surface heat flux has been reported by Pullepu et al. [44]. However, the assumption of the thin boundary layers and negligible curvature effects is implicit in almost all these analyses. Some of these ideas have also been extended to free convection in power-law fluids for axisymmetric shapes that can be used to recover approximate results for a slender cone. For instance, Acrivos [45] obtained the asymptotic solution of the boundary-layer equations for high values of the Prandtl number, which can be used to deduce the results for a slender cone. Subsequently, Shenoy [46] revisited this problem for power-law fluids and reported an expression for the Nusselt number for a slender cone based on the approximate integral method. Similarly, Marner and Kleppe [47] and Kleppe and Marner [48] obtained approximate solutions to the boundary-layer equations for the transient free and mixed convections to Bingham plastic fluids from a vertical isothermal plate. However, over the rather narrow range of conditions, they concluded that the values of the Nusselt number deviated by only 3% from the corresponding Newtonian results. The limited literature in this field has been reviewed in a recent book [49]. In addition to the preceding analytical studies, scant experimental results have been reported. For instance, Oosthuizen and Donaldson [50] used cones made of aluminum of varying heights and angles fitted with thermocouples to measure the temperature on the surface of the cone suspended in still air. Their results on the average Nusselt number span the range of conditions as the Grashof number (3.548×10^{7}–4.720×10^{8}), Prandtl number (0.7), and cone angles (3.5–11.5 deg). More recently, Ashjaee et al. [51] used the Mach–Zehnder interferometry technique to investigate the structure of the temperature field in the proximity of a cone. The results on the local and average Nusselt numbers in air over the surface of the cone are reported over the range of Rayleigh numbers (4.9×10^{5}–1.1×10^{6}) for a cone angle of 11.5 deg. It is worthwhile to add here that there have been very few studies about the momentum transfer (drag) and heat transfer from cones in the forced convection regime. Mishra et al. [13] have studied the laminar forced convection heat transfer from isothermal cones in Bingham plastic fluids. They reported the heat transfer to bear a positive dependence on the Reynolds, Prandtl, and Bingham numbers for both orientations: apex upward and downward. Similarly, the corresponding drag results for cones settling in power-law [52] and Bingham plastic [14] fluids have been reported recently, which are in line with the scant experimental results available in the literature. To the best of our knowledge, no other results are available on heat transfer from a cone in non-Newtonian fluids in forced, or free, or mixed convection regimes. Hence, this work aims to fill this gap in the literature, at least in the free convection heat transfer regime.

From the preceding discussion, it is thus fairly clear that most of the available results are based on the solution of the boundary-layer equations of natural convection, even in the case of Newtonian fluids. These analyses entail the assumptions of the high Grashof number and/or Prandtl numbers and negligible curvature effects. It is needless to add here that even less is known about the analogous problem in power-law and Bingham plastic fluids. Thus, the present work endeavours to fill this gap in the literature. In particular, numerical results based on the solution of the coupled momentum and energy equations for an isothermal cone in power-law and Bingham plastic fluids are reported here over wide ranges of parameters such as the Grashof number (10–10^4), the Prandtl number (5–100), the Bingham number (1–50), and the power-law index (0.4–1) but for a fixed value of cone angle of 60 deg. Detailed results on the structure of the flow and temperature fields in terms of the yielded and unyielded regions for Bingham plastic fluids, streamlines, isotherm contours, and the local Nusselt number are discussed to delineate the effect of each of these parameters. Included here are also the results for Newtonian fluids, especially at high Prandtl numbers. These serve as the baseline to draw qualitative inferences regarding the role of power-law and Bingham plastic fluid behaviors on free convection heat transfer from an isothermal cone. This paper is organized as follows: problem formulation along with the simplifying assumptions and the relevant dimensionless parameters used in this work are presented in Sec. II, followed by the description of the numerical solution methodology and the choice of numerical parameters used here in Sec. III. Section IV presents and discusses the new results obtained in this work, whereas the key findings are summarized in Sec. V.

II. Problem Formulation

Consider an isothermal cone (surface temperature of T_w) of diameter d and a cone angle ($\phi = 30$ deg) submerged in a quiescent fluid (power-law or Bingham plastic) at a temperature of $T_0 < T_w$ (Fig. 1). Evidently, the fluid density near the cone is minimum, and it increases linearly (Boussinesq approximation) with the temperature eventually attaining the value of ρ_0 corresponding to the reference temperature T_0. Such a density stratification gives rise to the upward flow of fluid, and heat transfer occurs by free (or natural) convection from the cone to the fluid or vice versa, depending upon the direction of the imposed temperature difference. The two configurations, the apex upward and downward of the cone, are considered here. Over the range of conditions encompassed here, the flow is assumed to be laminar, axisymmetric and incompressible, except for the body force term appearing in the z-momentum equation. To keep the level of complexity at a tractable level, the thermophysical properties of the fluid (heat capacity c_p, Bingham fluid viscosity μ_B, yield stress τ_0, thermal conductivity k, consistency index m, and power-law index n) are assumed to be temperature independent. This assumption is justified here because the maximum value of $\Delta T(= T_w - T_0)$ is only 5 K in this study. Furthermore, the viscous dissipation is also neglected here because the maximum value of the Brinkman number is on the order of 10^{-3}. Hence, the velocity and temperature fields are coupled only on account of the temperature-dependent density. For small values of $\Delta T(= T_w - T_0)$, the fluid density varies linearly with temperature via the Boussinesq approximation given as follows:

$$\rho_0 - \rho = \rho_0\beta(T - T_0) \tag{4}$$

In Eq. (4),

$$\beta = -\frac{1}{\rho}\frac{\partial\rho}{\partial T}\bigg|_p$$

is the coefficient of volumetric thermal expansion. Within the framework of these simplifying assumptions, the coupled equations

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

of motion and energy are written as follows (in their dimensionless forms):

Continuity equation:

$$\frac{1}{r}\frac{\partial(rU_r)}{\partial r}+\frac{\partial U_z}{\partial z}=0 \tag{5}$$

r-momentum equation:

$$\left(\frac{\partial U_r}{\partial t}+U_r\frac{\partial U_r}{\partial r}+U_z\frac{\partial U_r}{\partial z}\right)=-\frac{\partial p}{\partial r}+\frac{1}{\sqrt{Gr}}\left(\frac{1}{r}\frac{\partial(r\tau_{rr})}{\partial r}+\frac{\partial \tau_{zr}}{\partial z}\right) \tag{6}$$

z-momentum equation:

$$\left(\frac{\partial U_z}{\partial t}+U_r\frac{\partial U_z}{\partial r}+U_z\frac{\partial U_z}{\partial z}\right)=-\frac{\partial p}{\partial z}+\frac{1}{\sqrt{Gr}}\left(\frac{1}{r}\frac{\partial(r\tau_{rz})}{\partial r}+\frac{\partial \tau_{zz}}{\partial z}\right)+\Theta \tag{7}$$

Thermal energy equation:

$$\frac{\partial \Theta}{\partial t}+U_r\frac{\partial \Theta}{\partial r}+U_z\frac{\partial \Theta}{\partial z}=\frac{1}{\lambda}\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \Theta}{\partial r}\right)+\frac{\partial^2\Theta}{\partial z^2}\right) \tag{8}$$

where the values of the coefficients in Eqs. (6–8) are given in Table 1 for power-law and Bingham plastic fluids. Even though the major thrust of this work is on the steady flow regime, the transient terms are retained in Eqs. (6–8); and these are solved with a suitable time-integration scheme in order to ascertain the nature of the flow, i.e., whether the solution attains a steady state or not.

For Bingham plastic fluids, the deviatoric part of the stress tensor is given as follows [53]:

$$\tau_{ij}=\left(1+\frac{Bn}{\sqrt{\Pi_{\dot\gamma}}}\right)\dot\gamma_{ij}, \qquad \text{if } \Pi_\tau > Bn^2 \tag{9}$$

$$\dot\gamma_{ij}=0, \qquad \text{if } \Pi_\tau \le Bn^2 \tag{10}$$

Clearly, Eqs. (9) and (10) are inherently discontinuous and nondifferentiable. However, this difficulty is usually circumvented by employing a regularization method proposed by Papanastasiou [53], which provides a smooth transition from the solidlike to the fluidlike behavior. Based on the Papanastasiou modification, Eqs. (9) and (10) are rewritten as follows:

$$\tau_{ij}=\left(1+\frac{Bn\left[1-\exp\left(-M\sqrt{\Pi_{\dot\gamma}}\right)\right]}{\sqrt{\Pi_{\dot\gamma}}}\right)\dot\gamma_{ij} \tag{11}$$

where M is a regularization parameter. Much has been written about the other regularization methods and their merits and demerits in the literature [54]. Suffice it to add here that the two other commonly used regularization methods (namely, biviscous [55] and Bercovier and Engelman [56]) were also used in a few cases to corroborate the present predictions, as will be discussed later.

For power-law fluids, the deviatoric stress tensor is given as follows:

$$\tau_{ij}=m(\Pi_{\dot\gamma}/2)^{(n-1)/2}\dot\gamma_{ij} \tag{12}$$

Table 1 Coefficients in Eqs. (6–8)

Coefficient	Power-law fluid	Bingham plastic fluid
Gr	Gr_{PL}	Gr_{BP}
λ	$Pr_{\mathrm{PL}}\cdot Gr_{\mathrm{PL}}^{(1/n+1)}$	$Pr_{\mathrm{BP}}\cdot\sqrt{Gr_{\mathrm{BP}}}$

where $\dot\gamma_{ij}=(\nabla U+\nabla U^T)$. The aforementioned equations have been rendered dimensionless using d, $\sqrt{dg\beta\Delta T}$ and μ_B as the characteristic length, velocity, and viscosity scales, respectively, for Bingham plastic fluids. Thus, for instance, the rate of strain and extra stress components are scaled as $\sqrt{g\beta\Delta T/d}$ and $\mu_B(\sqrt{g\beta\Delta T/d})$, and the pressure is scaled as $\rho_0 dg\beta\Delta T$. The temperature is nondimensionalized as $\Theta=(T-T_0)/(T_w-T_0)$. Similarly, for power-law fluids, the corresponding characteristic scale for the viscosity is given as $m(\sqrt{g\beta\Delta T/d})^{n-1}$. The dimensional considerations suggest the velocity and temperature fields to be functions of three-dimensionless groups (namely, the Grashof number, Prandtl number, and Bingham number) or of the power-law index for a fixed cone angle and orientation of the cone. These dimensionless numbers are defined as follows:

Bingham number:

$$Bn=\frac{\tau_0}{\mu_B}\sqrt{\frac{d}{g\beta\Delta T}} \tag{13}$$

Grashof number for Bingham plastic fluids:

$$Gr_{\mathrm{BP}}=\frac{\rho^2 g\beta\Delta T d^3}{\mu_B^2} \tag{14}$$

Grashof number for power-law fluids:

$$Gr_{\mathrm{PL}}=\frac{\rho^2(g\beta\Delta T)^{2-n}d^{n+2}}{m^2} \tag{15}$$

The Prandtl number for a Bingham plastic fluid is given as follows:

$$Pr_{\mathrm{BP}}=\frac{c_p\mu_B}{k} \tag{16}$$

For power-law fluids, it is defined as follows:

$$Pr_{\mathrm{PL}}=\frac{\rho c_p}{k}\left(\frac{m}{\rho}\right)^{2/(1+n)}d^{(1-n)/(1+n)}(dg\beta\Delta T)^{(3(n-1)/2(n+1))} \tag{17}$$

Suffice it to say here that, in the limit of $Bn\to 0$ or $n\to 1$, these definitions coincide with that for Newtonian fluids. Finally, the no-slip ($U_r=U_z=0$) and the constant-temperature $\Theta=1$ boundary conditions are prescribed on the surface of the cone. On the other hand, the radial velocity and its gradient (i.e., $U_r=0, \partial U_r/\partial r=0$), as well as a uniform temperature ($\Theta=0$), have been used on the surface of the fictitious spherical envelope enclosing the cone. These conditions are similar to those used in the literature for Newtonian and power-law fluids [28,32] for spheres.

The aforementioned governing equations subject to these boundary conditions are solved numerically together with the two viscosity models in terms of the primitive variables (u–v–p–T). These are, in turn, postprocessed to obtain streamline and isotherm contours, the structures of the yielded and unyielded regions (for Bingham plastic fluids), and the local Nusselt number distribution around the surface of the cone.

The Nusselt number is defined as follows:

$$Nu_L=\frac{hd}{k}=-\left(\frac{\partial\Theta}{\partial n_s}\right) \tag{18}$$

where n_s is the outward drawn unit vector normal to the surface of the cone; finally, the surface averaged Nusselt number Nu_{avg} is obtained simply by integrating the local Nusselt number values over the surface of the cone. However, the singular point C (in Fig. 2) was treated by dual prescription, i.e., the value of local Nusselt number is calculated along the lines AB and BC and the two values are averaged. Because the grid used here is sufficiently fine, this approximation is reasonable here. Besides, in a few cases,

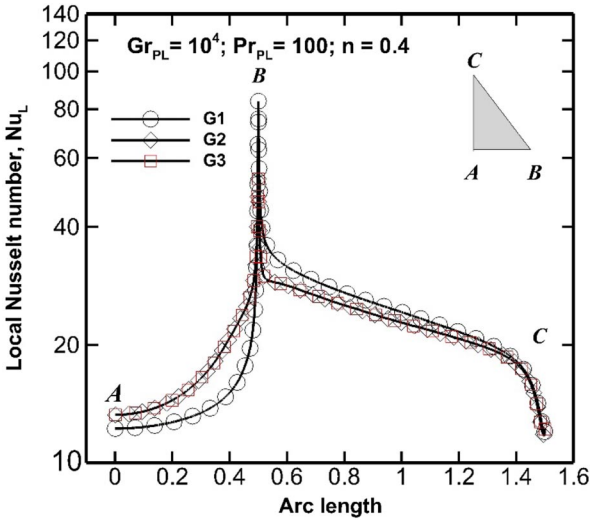


Fig. 2 Grid independence study at $Gr_{PL} = 10^4$, $n = 0.4$, and $Pr_{PL} = 100$.

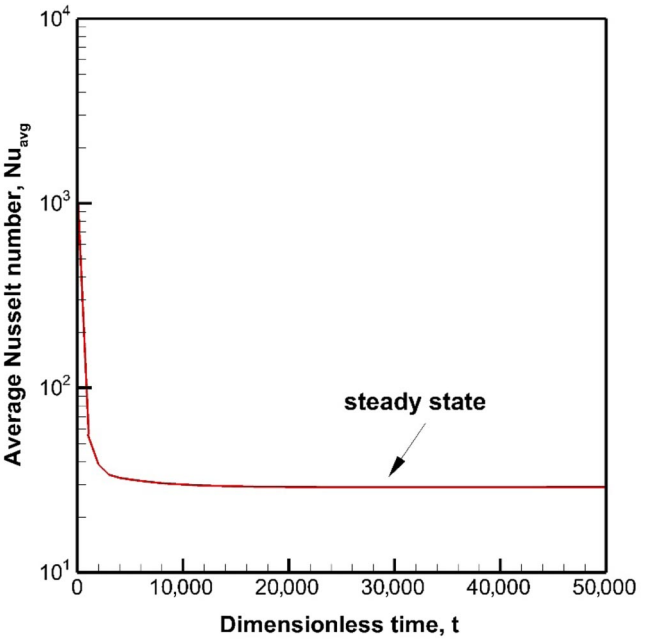


Fig. 3 Temporal evolution of the average Nusselt number at $Gr_{PL} = 10^4$, $n = 0.4$, and $Pr_{PL} = 100$ for apex upward orientation in a power-law fluid.

higher-order schemes were used to evaluate these values, and this influenced the resulting values negligibly.

Dimensional considerations of the problem studied here suggest the average Nusselt number to be a function of the Grashof number Gr, the Prandtl number Pr, the Bingham number Bn, or the power-law index n for a fixed cone angle and orientation; and this work endeavors to establish an understanding of this relationship.

III. Numerical Solution Methodology and Validation of Results

The governing equations [Eqs. (6–8)] together with the aforementioned boundary conditions for power-law and Bingham viscosity models are solved numerically using the finite element-based solver COMSOL Multiphysics (version 5.0). Owing to the boundary-layer formation on the surface of the cone, the velocity and temperature gradients are expected to be steep in this region; therefore, a fine mesh is used in this region. Similarly, the high mesh density is also used near the rim and apex of the cone, as well as near the yield surfaces separating the yielded and unyielded regions in the case of Bingham fluids. The computational domain is mapped with free triangular elements with nonuniform spacing. The axisymmetric, nonisothermal flow module has been employed to model the buoyancy driven free convective flow. The buoyant force term and the expression for viscosity for the two models are introduced via user-defined functions. The PARDISO direct solver is used to solve the resulting system of equations. To avoid the potential convergence difficulties at extreme values of the parameters, transient simulations were carried out to obtain the steady-state results as shown in Fig. 3. Initially, the approximate location of the yield surfaces was obtained, and then the grid was progressively refined manually in this region. The results for the lower values of the Bingham number Bn and the Grashof number Gr were used as initial estimates for this purpose. This entailed a degree of trial-and-error processes to obtain a satisfactory resolution of the yield surfaces separating the fluidlike and solidlike regions for given conditions. The backward differentiation formula for time stepping with $\Delta t = 0.1$ (dimensionless) is used. The criterion for a steady state is defined when the average Nusselt number levels off to a constant value in the $Nu_{avg} - t$ curve (Fig. 3), i.e., when the results change by less than 0.01% between the two successive steps. This approach also confirms that the flow is indeed steady over the range of conditions spanned in this study for Bingham and power-law fluids, except for one case. For the case of a cone with its apex pointing downward, for $Gr_{PL} = 10^4$, $Pr_{PL} = 100$, and $n = 0.4$, the velocity and temperature fields lost their steady nature along the base of the cone (Fig. 4; upper figure). This prompted us to explore the other values of the Grashof number at $n = 0.4$ and $Pr_{PL} = 100$ for this configuration. Indeed, the flow was found to be steady up to $Gr_{PL} = 0.9 \times 10^4$ at $n = 0.4$ and $Pr_{PL} = 100$ for this orientation of the cone. In all other cases, the use of a time-dependent solver accelerated the rate of convergence to the final steady-state solution [57]. A relative stopping criterion of 10^{-5} for the equations of motion and energy is employed, and any further reduction in this value had a negligible effect on the results.

A systematic grid and domain independence study has been carried out here in order to ensure that the present results are free from numerical artefacts. To approximate an unconfined flow, an artificial spherical computational domain of radius R_∞ (see Fig. 1b) has been used here with a sufficiently large value of R_∞. For this purpose, three different values of $R_\infty = 100$, 200, and 300 were tested; and $R_\infty = 200$ was found to be adequate for the present results to be free from boundary effects. Two comments are in order here. First, the natural convection flowfield can be influenced by the shape of the domain, especially when a relatively small computational domain is used [1]. Because, in this study, the radius of the computational domain R_∞ is 100 times the radius of the cone, the shape of the domain is unlikely to influence the results. Second, depending upon the orientation of the cone, the region above or below the apex is of interest, and so the computational domain is centered at the apex and due to the large value of R_∞ used here; this choice also does not influence the solution. Similarly, grid independence studies were carried out for both power-law and Bingham fluids at Gr_{PL} or $Gr_{BP} = 10^4$, $n = 0.4$ or $Bn = 1$, and Pr_{PL} or $Pr_{BP} = 100$ by employing three grids, namely, $G1$, $G2$, and $G3$ consisting of 200, 675, and 1350 elements, respectively, on the surface of cone. The resulting local Nusselt number values along the surface of the cone are shown in Fig. 3 for a power-law fluid in which the predictions of $G2$ and $G3$ are within $\pm 0.01\%$ each other; therefore, grid $G2$ is considered to be adequate over the range of conditions spanned here. The influence of the regularization parameter M on the yield surfaces (Fig. 5a) and the local Nusselt number was monitored, and the value of $M = 10^8$ was found to be sufficient. Also, the fluidlike (yielded) and solidlike (unyielded) regions for Bingham plastic fluids were delineated by employing the von Mises criterion with the relative tolerance of 10^{-5} [58]. Furthermore, Fig. 5b shows a typical comparison between the predictions of the yield surfaces obtained using the three commonly employed regularization methods mentioned earlier. Indeed, the three predictions are seen to be in near perfect agreement. (In Fig. 5b, PP denotes the Papanastasiou model ($M = 10^8$) [53], BV denotes the biviscous model ($\mu_Y = 10^7 \mu_B$), and BE denotes the Bercovier

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

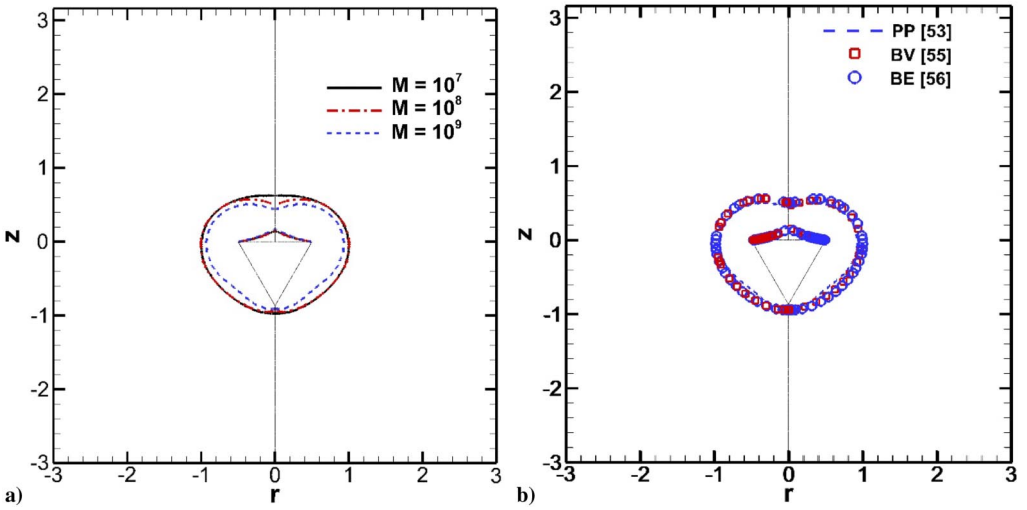


Fig. 4 Distribution of the local Nusselt number along the surface of the cone at $Pr_{PL} = 100$ in power-law fluids.

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

and Engelman model ($\varepsilon = 10^{-9}$) [56].) Next, Table 2 compares the resulting values of drag coefficient C_D and the average Nusselt number; once again, an excellent match is seen to exist here, thereby inspiring confidence in the validity of the regularization method [namely, Eq. (11)] used in this work. In summary, the results reported herein are based on $R_\infty = 200$, grid $G2$, and $M = 10^8$. Further support for this selection is provided in the next paragraph in which a few benchmark comparisons are presented.

First, Fig. 6 shows a comparison between the present results with that based on the boundary-layer analysis for air [34,38]. As expected, the correspondence improves with the increasing Grashof number. This is not at all surprising because the assumptions of the thin boundary layer and negligible curvature effects inherent in the boundary-layer analysis are valid only at high Grashof numbers. Similarly, the temperature profile on the lateral surface of the cone along the normal direction is shown in Fig. 7 at $Gr_L^* = 5 \times 10^6$; an excellent match between the present results and that of Hering and Grosh [34] is seen to exist here also. Furthermore, Table 3 compares the present results with the predictions of Eq. (2), due to Stewart [37]; once again, the gap between the two results progressively decreases with the increasing Grashof number, which is in line with the assumptions inherent in the boundary-layer analysis leading to Eq. (2). A typical comparison for power-law fluids is shown in Table 4 at high Grashof and Prandtl numbers. All three predictions are seen to be in excellent agreement. Based on the preceding benchmark comparisons coupled with the recommendation of Mitsoulis and Tsamopolous [54], the results reported herein are based on the application of Eq. (11), due to Papanastasiou [53]. Also, the preceding validations coupled with the extensive benchmark comparisons for a sphere, spheroid, disk, etc., presented elsewhere [28–30] lend credibility to the precision and reliability of the present numerical results.

IV. Results and Discussion

A. Streamlines and Isotherm Contours

As noted earlier, the density of the fluid near the heated cone will be lower than that of the fluid in the far field. Therefore, the hot fluid rises and is replaced by cold fluid from the sides and/or from beneath the heated cone, depending upon the strength of the buoyancy-induced flow (i.e., Grashof number). This, in turn, sets up an upward current leading to the formation of a plume above the heated cone. Clearly, the flowfield around the cone is expected to be the strongest. Representative streamline and isotherm contours are shown in Fig. 8 for scores of values of the shear-thinning index and the two extreme values of Grashof number. Clearly, at low Grashof numbers, cold fluid is entrained from beneath the cone, whereas the rising fluid is replenished from sides at high Grashof numbers for both configurations. No flow detachment was seen to occur over the range of conditions considered here. The corresponding isotherm contours are shown on the left of the Fig. 8. As expected, the temperature gradient sharpens with the increasing Grashof number and decreasing power-law index due to progressive thinning of the thermal boundary layer [16,17]. Thus, the shear-thinning fluid behavior is expected to enhance the rate of heat transfer.

Figure 9 shows the corresponding results for Bingham fluids for a range of values of the Bingham number. For a fixed value of the Bingham number, the momentum and thermal boundary layers

Table 2 Comparison of drag coefficient and Nusselt number predicted from various regularization models: $Bn = 5$; $Gr = 10^4$; $Pr = 100$ (apex upward)

	Papanastasiou [53]		Biviscous model [55]		Bercovier and Engelman [56]	
Pr	C_D	Nu	C_D	Nu	C_D	Nu
5	1.1805	5.1893	1.1804	5.1913	1.1806	5.1811
100	1.0629	5.6274	1.0629	5.6302	1.0599	5.6282

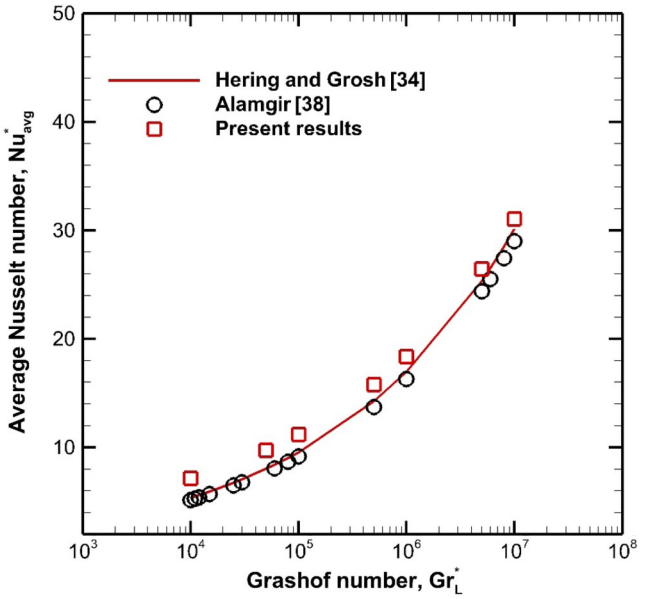


Fig. 5 Representations of a) effect of regularization parameter M on yield surface at $Gr_{BP} = 10^4$, $Bn = 20$, and $Pr_{BP} = 100$; and b) comparison between prediction of three regularization methods: PP, BV, and BE.

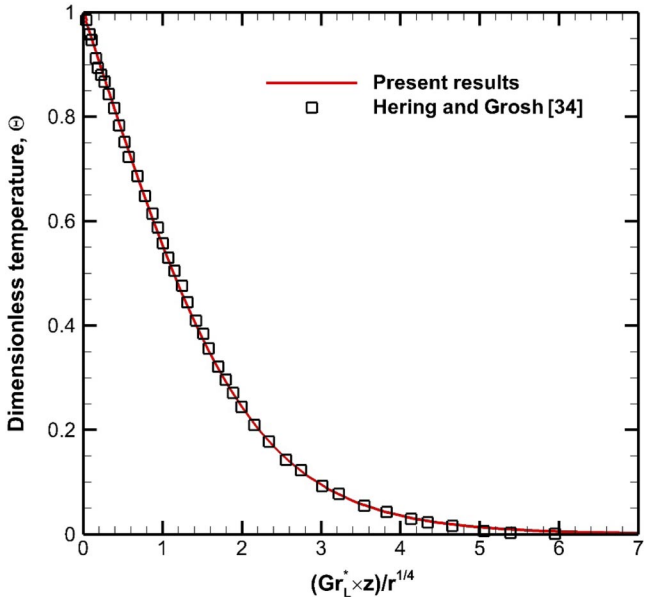


Fig. 6 Comparison of the average Nusselt number Nu^*_{avg} over the slant area of the right circular cone ($\phi = 30$ deg) with Hering and Grosh [34] and Alamgir [38] ($Pr = 0.7$; and $Gr^*_L = Gr_L \times \cos\phi$).

progressively thin with the increasing Grashof number, although this effect is the strongest in Newtonian fluids, i.e., $Bn = 0$. This is not at all surprising because the increasing Bingham number (yield stress) suppresses the propensity for fluid deformation. This, in turn, suggests that no shearing will occur in the limit of $Bn \to \infty$. Such a limiting behavior can also be inferred from the isotherms close to the cone, where the temperature contours are seen to be diffusing radially (at $Bn = 50$), which is akin to that in the case of pure conduction, irrespective of the orientation of the cone. This is in stark contrast to the behavior of Newtonian and power-law fluids, in which this flow regime is observed only in the limit of $Gr \to 0$.

In summary, the velocity and temperature gradients in the thin boundary layers sharpen with the increasing Grashof number and decreasing value of the power-law index. On the other hand, with the increasing Bingham number, the contribution of conduction to the overall heat transfer increases, eventually approaching the pure conduction limit.

Table 3 Comparison of the present results with that of Stewart [37] at $Pr = 100$ and $\phi = 40$ deg in Newtonian fluids (apex upward case)

Gr_L	Nu^*_{avg} (present)	Nu^*_{avg} [Eq. (2)]	Difference, %
10^3	6.64	7.56	13.90
10^4	11.81	12.32	4.35
10^5	21.00	21.15	0.75

Table 4 Comparison of present results with the boundary-layer solution reported by Shenoy [49] and Acrivos [45] in power-law fluid ($Gr_L = 10^7$; $Pr = 10^4$; $\phi = 5$ deg):[a] ($Nu^*_{avg}/Gr_L^{1/2(n+1)} Pr^{n/(3n+1)}$)

n	Shenoy [49]	Acrivos [45]	Present results
0.5	0.623	0.650	0.629
1	0.718	0.710	0.690
1.5	0.750	0.767	0.773

[a]Here, all dimensionless numbers are defined based on the slant height of the cone as a characteristic length scale.

B. Structure of Yielded and Unyielded Regions

As mentioned earlier, the key distinguishing feature of the Bingham plastic fluids is the simultaneous coexistence of the yielded and unyielded regions in the flow domain determined by the magnitude of the local stress tensor vis-à-vis the value of the yield stress. Clearly, the unyielded zones are undesirable both from homogenization and heating/cooling standpoints. Overall, it has a deleterious influence on the product homogeneity and quality. Therefore, it is useful to examine the influence of the Bingham number, the Grashof number, and the Prandtl number on the shape and size of the yielded and unyielded regions as shown in Fig. 10. A detailed inspection of these results suggests that there are three possibilities in this regard, depending upon the value of the Bingham number. At low Bingham numbers (such as at $Bn = 5$), the flow domain is covered with more yielded regions than the unyielded regions, which is similar to the case of a Newtonian fluid. The unyielded regions in this regime are in the form of small patches along the z axis above and below the cone and caplike structures attached to the base of the cone, in both orientations, and regardless of the value of the Prandtl number. Thus, convection occurs in the bulk of the fluid under these conditions. For intermediate values of Bingham numbers (as at $Bn = 20$), the yielded region is mainly confined to a cavity encapsulating the cone, whereas the material away from the cone remains frozen like a solid. In this regime, both conduction and convection contribute to the overall heat transfer in varying proportions. Furthermore, there are two unyielded regions observed in the intermediate range of Bingham number, namely, a caplike structure adhering to the base of the cone in both orientations and an envelope of the unyielded fluid, as can be seen in Fig. 10 at $Bn = 20$. The caplike structure is static in nature. Finally, at very high Bingham numbers ($Bn > 20$), there is a very small region of yielded fluid. Obviously, conduction dominates the overall heat transfer in this regime. To corroborate the predictions of the yielded surfaces obtained using Eq. (11), limited results were also obtained using another regularization scheme (namely, the biviscous model [55]), and the two predictions are generally seen to be in agreement with each other, except for the minor differences in the boundary of the fluidlike cavity encapsulating the cone. Such minor differences, however, exert virtually no influence on the value of the Nusselt number, which is determined primarily by the value of the temperature gradient on the surface of the cone (Table 2). Included in Table 2 are also the predictions of the regularization method of Bercovier and Engelman [56]. All these predictions are seen to be in very good agreement.

C. Distribution of Local Nusselt Number

The local Nusselt number denotes the nondimensional rate of heat transfer from the cone to the fluid. For a power-law fluid, typical distribution of the local Nusselt number around the cone is shown in Fig. 4. Evidently, for the apex downward configuration, the maximum value occurs at the apex (point A), which progressively decreases along the lateral surface in a manner similar to that in the boundary layer developed along a plane surface; and the minimum heat transfer is seen to occur at point C, irrespective of the values of the Grashof number or the Prandtl number and power-law index. However, the behavior for $n = 0.4$ is seen to be different from that for the other values of $n > 0.4$, especially along surface BC. This is due to the loss of steadiness of the flow at $Gr_{PL} = 10^4$ and $Pr_{PL} = 100$ at $n = 0.4$. In this case, the flowfield neither converged to a steady state nor became fully periodic. Thus, additional simulations were carried out at $Gr_{PL} = 0.9 \times 10^4$ and $Gr_{PL} = 0.95 \times 10^4$ at these values of $Pr_{PL} = 100$ and $n = 0.4$. Indeed, a steady-state solution was obtained at $Gr_{PL} = 0.9 \times 10^4$ and not at $Gr_{PL} = 0.95 \times 10^4$, thereby suggesting the critical values of $Gr_{PL} = 0.9 \times 10^4$ for the steady flow regime at these values of n and Pr_{PL}. Further examination of the flowfield revealed the formation of a small wake along BC, which did not initiate at B or extend all the way up to C. Thus, the unusual increase in the local Nusselt number seen in Fig. 4 can be ascribed to the presence of such a wake and, because it only spans a part of BC, the local

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

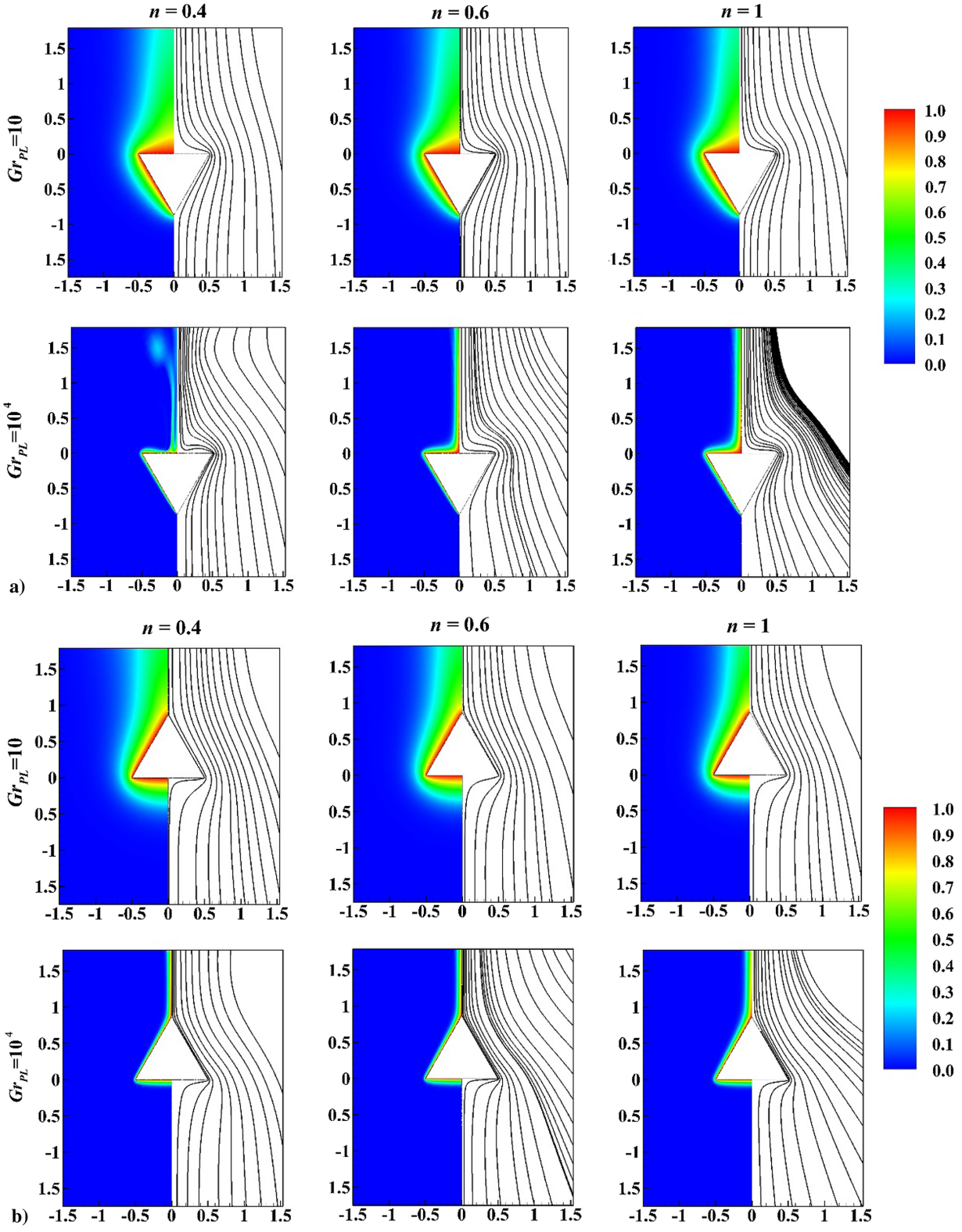


Fig. 7 Comparison of temperature profile along line perpendicular to slant surface of cone with that of Hering and Grosh [34] at $Gr_L^* = 5 \times 10^6$ and $Pr = 0.7$ with apex upward configuration ($\phi = 30$ deg).

Nusselt number drops after showing an upward trend. However, such a weak transient nature of the local Nusselt number does not impact the average value of the Nusselt number. On the other hand, for the apex upward configuration, at low Grashof numbers (i.e., $Gr_{\mathrm{PL}} = 10$), the maximum rate of heat transfer occurs at the apex (point C), irrespective of the value of the Prandtl number and power-law index, whereas for high Grashof numbers (i.e., $Gr_{\mathrm{PL}} = 10^4$), the maximum value occurs at point B and the minimum rate of heat transfer is observed at the front stagnation point (point A). In this case, the velocity and temperature fields were found to be steady over the range of conditions spanned in this work. The local Nusselt number varies very little along AB because there is no component of gravity along this surface and heat transfer must occur mainly by diffusion slightly modified by weak

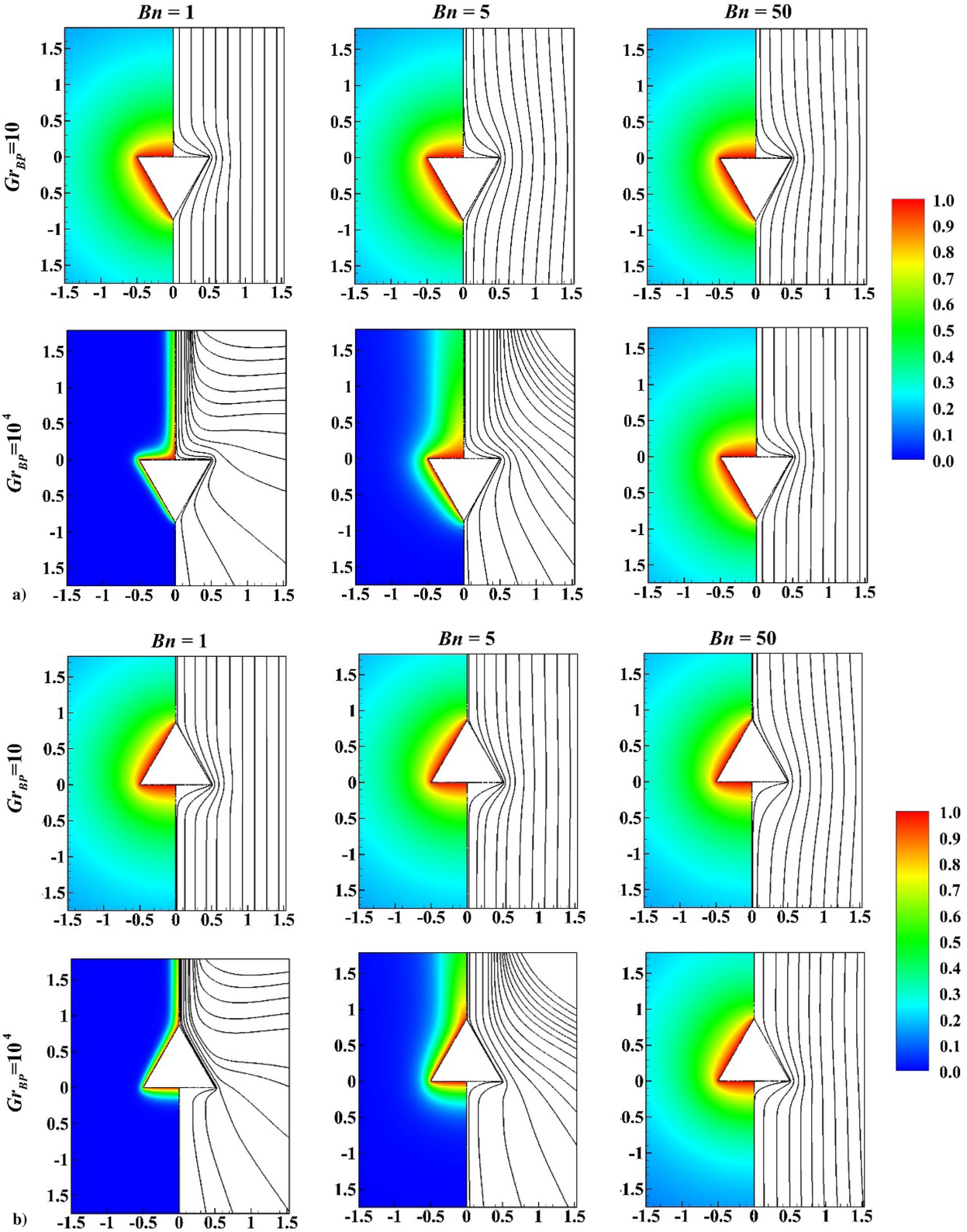


Fig. 8 Representative streamline (right half) and isotherm contours (left half) for power-law fluids at $Pr_{PL} = 100$: a) apex downward, and b) apex upward.

flow, eventually suddenly accelerating at B. In the opposite configuration, a fluid element experiences sudden deceleration at point B, thereby leading to the formation of a small wake and/or time effects. For the sake of comparison, the corresponding results for pure conduction are also included in these figures. At $Gr_{PL} = 10$ (not shown in Fig. 4), there is a very weak advection because the values of the local Nusselt number are pretty close to that for a pure conduction. The two values, however, differ significantly from each other at $Gr_{PL} = 10^4$.

Similarly, for a Bingham plastic fluid, the typical variation of the local Nusselt number along the surface of the cone is shown in Fig. 11 for a range of values of the Bingham number. The overall trend of the local Nusselt number is similar to that for a power-law fluid. As noted in the previous section, the heat transfer should decrease with

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

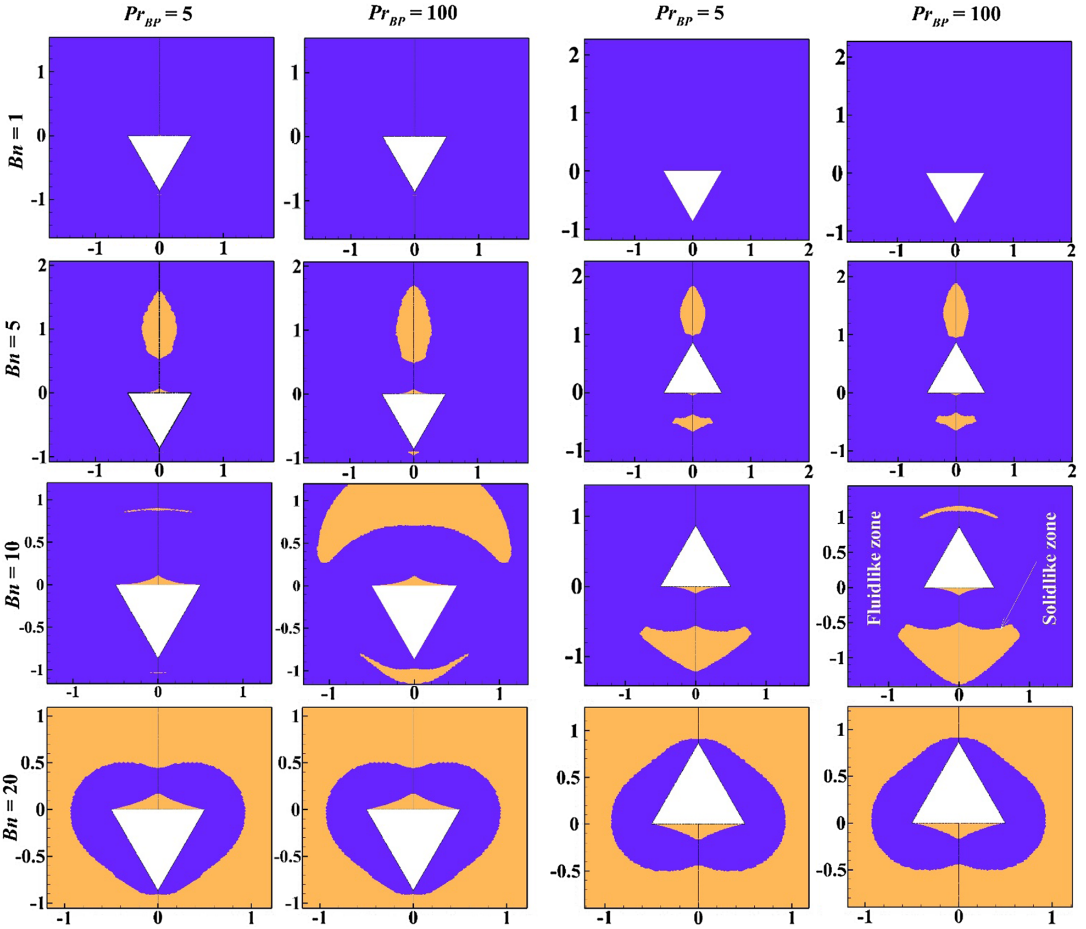


Fig. 9 Representative streamline (right half) and isotherm contours (left half) for Bingham fluids at $Pr_{PL} = 100$: a) apex downward, and b) apex upward.

increasing Bingham number, as can be clearly seen from Fig. 11. This is simply due to the formation of caplike structures around the base of the cone, which are observed in both orientations. In summary, due to the formation of the unyielded regions at the base of the cone, heat transfer in these regions is impeded in Bingham plastic fluids with reference to that in Newtonian fluids, thereby lowering the overall heat transfer.

D. Average Nusselt Number

Notwithstanding the fact that the detailed flow patterns (streamline, isotherm, yield surfaces) and distribution of the local Nusselt number provide information about the so-called "cold" and "hot" spots in the flow domain (relative contributions of various parts of the cone to the overall heat transfer, etc.), it is the surface averaged value of the Nusselt number that is required to size the process heat exchanging equipment. Dimensional considerations suggest the average Nusselt number to be a function of the Grashof number, the Prandtl number, and the Bingham number or power-law index for a fixed orientation and cone angle.

For a power-law fluid, this relationship is written as follows:

$$Nu_{\text{avg}} = f(Gr_{\text{PL}}, Pr_{\text{PL}}, n) \tag{19}$$

and, for a Bingham plastic fluid,

$$Nu_{\text{avg}} = f(Gr_{\text{BP}}, Pr_{\text{BP}}, Bn) \tag{20}$$

Figure 12 shows representative results elucidating the influence of Gr_{PL}, Pr_{PL}, and n for the two orientations on the average Nusselt number in power-law fluids. Although the average Nusselt number bears a positive dependence on both the Grashof and Prandtl numbers, the Nusselt number is seen to be slightly (up to 14%) larger for the cone with its apex facing down than that for the apex pointing upward. In the later orientation, when the fluid strikes the base of the cone, the fluid motion along the base must occur by diffusion before the gravity comes into action at the rim of the cone. Clearly, the Nusselt number increases with the decreasing power-law index, but only at high values of the Grashof number when a degree of advection is present. At low Grashof numbers, the shear-thinning behavior is seen to impede heat transfer, although the effect is very small. It is useful to recall here that the average Nusselt scales as $Nu_{\text{avg}} \sim Ra^{1/4}$ in Newtonian fluids, i.e., the scaling is identical with respect to the Grashof and Prandtl numbers. However, in the case of power-law fluids, this scaling is modulated by the value of the power-law index [45,46,59]. This feature is adequately captured by introducing the following modified form of the Rayleigh number Ω as follows [59]:

$$\Omega = Gr_{\text{PL}}^{1/(2(n+1))} Pr_{\text{PL}}^{n/(3n+1)} \tag{21}$$

In the limit of Newtonian fluid behavior ($n = 1$), Ω reduces to $Ra^{1/4}$ as expected. On the other hand, for $n = 0.4$, it predicts $\Omega \sim Gr_{\text{PL}}^{0.36} Pr_{\text{PL}}^{0.18}$, i.e., a stronger role of the Grashof number and the other way around for the Prandtl number than that in Newtonian fluids. Following the success of Ω in correlating experimental data for spheres [59] and numerical results for spheres and spheroids [24,25], the present results are best represented by the following expression:

$$Nu_{\text{avg}} = 2.205 + a(\Omega)^b \quad \text{for } \Omega \le 10 \tag{22}$$

$$Nu_{\text{avg}} = a(\Omega)^b \quad \text{for } \Omega > 10 \tag{23}$$

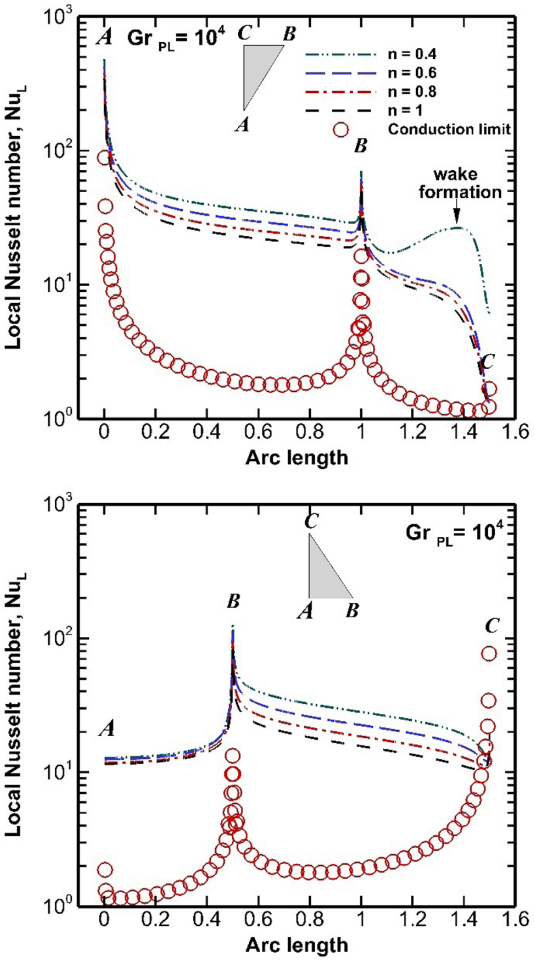


Fig. 10 Structure of yielded and unyielded regions at $Gr_{BP} = 10^4$.

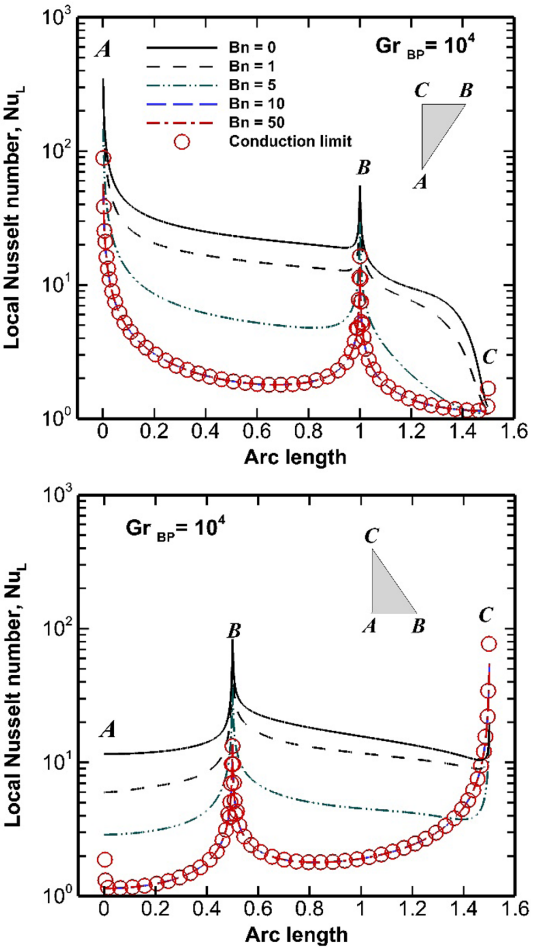


Fig. 11 Distribution of the local Nusselt number along the surface of the cone at $Pr_{BP} = 100$ in Bingham plastic fluids.

Clearly, in the limit of pure conduction (i.e., $\Omega \to 0$), Eq. (22) approaches the conduction limit of $Nu_{avg} = 2.205$ for both orientations. The values of the constants appearing in Eqs. (22) and (23) (based on 300 numerical data points for each configuration) along with the mean and maximum errors for the two configurations are included in Table 5. Figure 13 shows the parity plot; evidently, the influence of the cone orientation only begins to manifest at about $\Omega > \sim$10–15 or so, whereas the conduction limit is seen to be approached at about $\Omega \sim 0.5$. Also, the effect of power-law index is naturally built into the definition of Ω here.

For a Bingham plastic fluid, the representative functional dependence of the average Nusselt number on Pr_{BP} and the Bingham number Bn is shown in Fig. 14 for $Gr_{BP} = 10^4$. At low Grashof numbers, the average Nusselt number is very close (within ~1%) to the value for the conduction limit due to the formation of the unyielded regions. At high Grashof numbers, the average Nusselt number progressively decreases from its maximum value at $Bn = 0$ (Newtonian fluid) to the conduction limit at $Bn \geq Bn_{max}$. Such dependence of the average Nusselt number on the Bingham number is in stark contrast to that seen in the forced convection regime, as noted earlier [13]. The present numerical data is correlated in the following form:

$$Nu_{avg} = 2.205 + a^*(Gr^*_{BP}Pr^*_{BP})^{1/4}\left(1 - \frac{Bn^*}{0.15\sqrt{Gr^*_{BP}}}\right)^{b^*}$$

$$\text{for } Bn^* > 0.15\sqrt{Gr^*_{BP}} \tag{24a}$$

$$Nu_{avg} = 2.205 \quad \text{for } Bn^* \leq 0.15\sqrt{Gr^*_{BP}} \tag{24b}$$

where the modified definitions of the Grashof number, the Prandtl number, and the Bingham number are defined based on rescaling the viscosity as $\eta_{eff} \sim \mu_B + \tau_0/\sqrt{g\beta\Delta T/d}$ as opposed to μ_B, and these are given as follows [60]:

$$Gr^*_{BP} = \frac{Gr_{BP}}{(1+Bn)^2} \tag{25}$$

$$Pr^*_{BP} = Pr_{BP}(1+Bn) \tag{26}$$

$$Bn^* = \frac{Bn}{(1+Bn)} \tag{27}$$

The values of the correlation constants a^* and b^* for the two orientations of the cone are presented in Table 6. Equation (24) is based on 700 numerical data points over the following ranges of conditions: Grashof number $10 \leq Gr_{BP} \leq 10^4$, Prandtl number $5 \leq Pr_{BP} \leq 100$, and Bingham number $1 \leq Bn \leq 50$.

It is obvious that the flowfield in Bingham fluids is determined by a complex interplay between the buoyancy-induced inertial stress and viscous stress (including the effect of yield stress). For a fixed value of the Prandtl number and orientation, the ratio of the yield stress and buoyancy-induced stress can be represented in terms of $Bn/\sqrt{Gr_{BP}}$, which has been found to collapse the data onto a single curve. Following this reasoning [30,31,61], the average Nusselt number is plotted versus $Bn/\sqrt{Gr_{BP}}$ in Fig. 15, and the critical value of $Bn/\sqrt{Gr_{BP}}$ is seen to be ≈0.15 for both orientations; beyond which, the buoyancy-induced stress is insufficient to overcome the yield

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

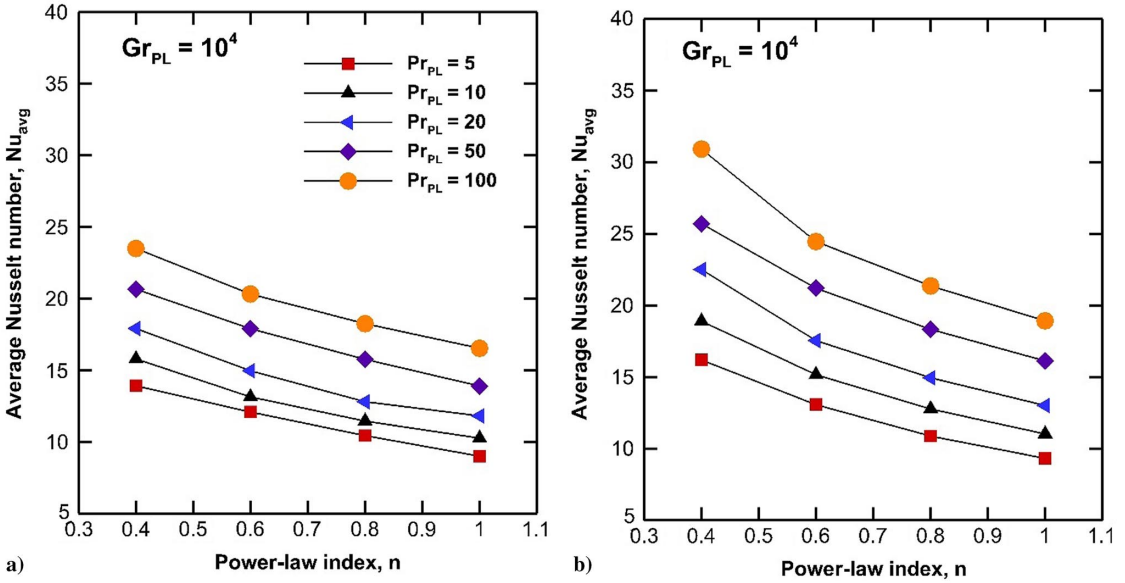


Fig. 12 Functional dependence of the average Nusselt number on the Prandtl number and power-law index at $Gr_{PL} = 10^4$: a) apex upward, and b) apex downward.

Table 5 Values of the correlation constants in Eqs. (22) and (23)

Range	a	b	Average error, %	Maximum error, %
		Apex downward		
$\Omega \leq 10$	0.49	1.01	3.36	9.59
$\Omega > 10$	1.03	0.81	5.32	16.63
		Apex upward		
$\Omega \leq 10$	0.48	1.01	2.87	8.24
$\Omega > 10$	1.33	0.70	5.60	19.25

stress to induce any convection. In other words, irrespective of the value of the Prandtl number and orientation, one can use the conduction value of the Nusselt number for $Bn \geq 0.15\sqrt{Gr_{BP}}$, which is equivalent to $Bn^* \geq 0.15\sqrt{Gr^*_{BP}}$.

Before leaving this section, it is useful to reiterate the key assumptions made in this work. The two main assumptions implicit in this study include the constant thermophysical properties (except the fluid density) and the negligible viscous dissipation. Among all the relevant properties, fluid viscosity is perhaps the most temperature-sensitive one. In the context of power-law fluids, although the power-law index is relatively insensitive to modest temperature changes, the consistency coefficient m exhibits the usual Arrhenius-type dependence [16,17]. Similarly, in the case of Bingham plastic fluids, although the Bingham viscosity follows the Arrhenius behavior [16], the effect of the temperature on the fluid yield stress τ_0 is much less clear [16]. The second assumption concerns the negligible viscous dissipation (small values of the Brinkman number). In principle, there is no difficulty in incorporating both these effects in the analysis, except that the coupling between the flow and the temperature fields would be accentuated, which may lead to potential convergence problems. Both these aspects will be explored in future studies in this field.

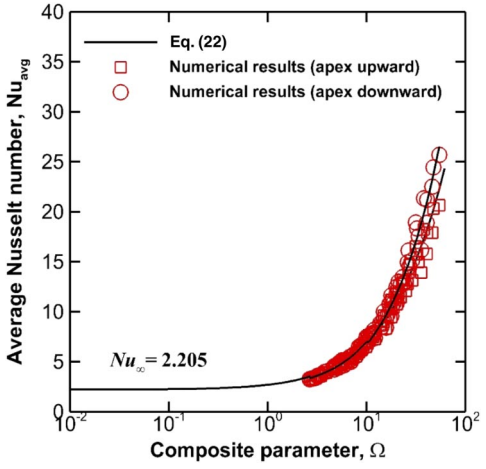


Fig. 13 Effect of the composite parameter Ω on the average Nusselt number.

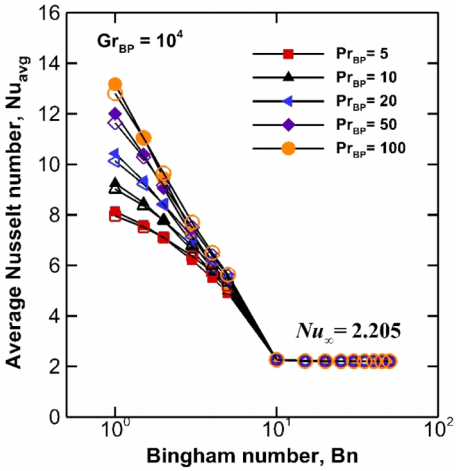


Fig. 14 Variation of the average Nusselt number and Prandtl number in a Bingham plastic fluid (filled symbols: apex upward configuration; unfilled symbols: apex downward configuration).

Table 6 Values of the correlation constants in Eq. (24)

	a^*	b^*	Average error, %	Maximum error, %
Apex downward	0.52	2.68	6.09	20.73
Apex upward	0.49	2.16	7.60	33.58

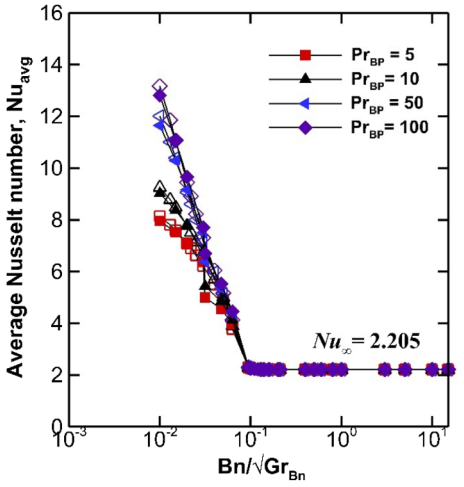


Fig. 15 Dependence of the average Nusselt number on the composite parameter ($Bn/\sqrt{Gr_{BP}}$) in Bingham fluids (filled symbols: apex upward configuration; unfilled symbols: apex downward configuration).

V. Conclusions

In this work, laminar natural convection from an isothermal cone in power-law and Bingham fluids has been investigated. Extensive results are reported over the range of conditions as follows: Grashof number (10–10^4); Prandtl number (5–100); Bingham number (1–50); and power-law index (0.4–1). The key findings of this work can be summarized as follows:

1) The phenomenon of free convection in power-law fluids is qualitative similar to that in Newtonian fluids, except for the fact that the influence of Grashof and Prandtl numbers on the Nusselt number is modulated by the value of the power-law index.

2) Broadly, shear-thinning fluid behavior promotes heat transfer over and above that in Newtonian fluids at fixed values of the Grashof and Prandtl numbers as long as there is a modest degree of convection.

3) In contrast, the fluid yield stress has deleterious effect on the overall heat transfer due to the unsheared blobs of fluid adhering to the surface of the cone. This is in contrast to its role in the forced convection regime, in which the average Nusselt number bears a positive dependence on the Bingham number.

4) In general, for fixed values of Gr_{BP} and Pr_{BP}, the average Nusselt number progressively decreases from its maximum value in a Newtonian fluid ($Bn = 0$) to the conduction limit at a critical value of the Bingham number.

5) A criterion for the onset of free convection in Bingham plastic fluids is presented here.

6) Finally, predictive correlations for both power-law and Bingham plastic fluids have been formulated that can be used to interpolate the present results and/or to estimate the value of the Nusselt number in a new application.

Acknowledgments

R. P. Chhabra would like to gratefully acknowledge the award of a J. C. Bose fellowship (by the Department of Science and Technology, Government of India, New Delhi, reference number SR/S2/JCB-06/2014) to him for the period 2015–2020.

References

[1] Martynenko, O. G., and Khramtsov, P. P., *Free-Convective Heat Transfer: With Many Photographs of Flows and Heat Exchange*, Springer, Berlin, 2005.

[2] Oosthuizen, P. H., and Kalendar, A., *Natural Convective Heat Transfer from Horizontal and Near Horizontal Surfaces*, Springer, New York, 2018.

[3] Kalendar, A., Karar, S., Kalendar, A., and Oosthuizen, P., "Correlations for Natural Convective Heat Transfer from Isothermal Surface of Octagonal and Hexagonal Shapes of Different Aspect Ratios," *Heat Transfer: Asian Research*, Vol. 46, No. 4, 2017, pp. 362–383. doi:10.1002/htj.21219

[4] Lee, S., Yovanovich, M., and Jafarpur, K., "Effects of Geometry and Orientation on Laminar Natural Convection from Isothermal Bodies," *Journal of Thermophysics and Heat Transfer*, Vol. 5, No. 2, 1991, pp. 208–216. doi:10.2514/3.249

[5] Yovanovich, M. M., "On the Effect of Shape, Aspect Ratio and Orientation Upon Natural Convection from Isothermal Bodies of Complex Shape," *ASME National Heat Transfer Conference*, Vol. 82, American Soc. of Mechanical Engineers, Fairfield, NJ, Dec. 1987, pp. 121–129.

[6] Yovanovich, M., "Natural Convection from Isothermal Convex Bodies-Simple Models for Bounds on Body-Gravity Function," *34th Thermophysics Conference*, AIAA Paper 2000-2581, June 2000. doi:10.2514/6.2000-2581

[7] Eslami, M., and Jafarpur, K., "Laminar Free Convection Heat Transfer from Isothermal Convex Bodies of Arbitrary Shape: A New Dynamic Model," *Heat and Mass Transfer*, Vol. 48, No. 2, 2012, pp. 301–315. doi:10.1007/s00231-011-0885-6

[8] Arabi, P., and Jafarpur, K., "Effect of Different Flow Regimes on Free Convection Heat Transfer from Isothermal Convex Bodies over all Range of Rayleigh and Prandtl Numbers," *Heat and Mass Transfer*, Vol. 52, No. 8, 2015, pp. 1665–1682. doi:10.1007/s00231-015-1683-3

[9] Dixon, B. D., and Parekh, J. V., "Use of the Cone Penetrometer for Testing the Firmness of Butter," *Journal of Texture Studies*, Vol. 10, No. 4, 1980, pp. 421–434. doi:10.1111/j.1745-4603.1980.tb00868.x

[10] Kuang, B., Mahmood, H. S., Quraishi, M. Z., Hoogmoed, W. B., Mouazen, A. M., and Van Henten, E. J.," "Sensing Soil Properties in the Laboratory, In Situ, and On-Line: A Review," *Advances in Agronomy*, edited by D.L. Spark, Academic Press, Cambridge, England, U.K., 2012, pp. 155–223.

[11] Tiwari, A. K., and Chhabra, R. P., "Laminar Free Convection in Bingham Plastic Fluids from an Isothermal Semicircular Cylinder," *Journal of Thermophysics and Heat Transfer*, Vol. 30, No. 2, 2016, pp. 369–378. doi:10.2514/1.T4688

[12] Patel, S. A., and Chhabra, R. P., "Laminar Free Convection in Bingham Plastic Fluids from an Isothermal Elliptic Cylinder," *Journal of Thermophysics and Heat Transfer*, Vol. 30, No. 1, 2016, pp. 152–167. doi:10.2514/1.T4578

[13] Mishra, P., Tiwari, A. K., and Chhabra, R. P., "Effect of Orientation on Forced Convection Heat Transfer from a Heated Cone in Bingham Plastic Fluids," *International Communications in Heat and Mass Transfer*, Vol. 93, No. 1, 2018, pp. 34–40. doi:10.1016/j.icheatmasstransfer.2018.02.018

[14] Mishra, P., Tiwari, A. K., and Chhabra, R. P., "Effect of Orientation on the Drag of a Cone Settling in a Bingham Plastic Fluid," *Particuology*, Vol. 43, April 2019, pp. 157–170. doi:10.1016/j.partic.2018.02.004

[15] Mishra, P., Gupta, S., and Chhabra, R.P., "Effect of Orientation on Drag Characteristics of a Cone Settling in Power-Law Fluids," *Journal of Chemical Engineering of Japan*, Vol. 52, No. 1, 2019, pp. 19–30.

[16] Chhabra, R. P., *Bubbles, Drops and Particles in Non-Newtonian Fluids*, 2nd ed., CRC Press, Boca Raton, FL, 2006.

[17] Chhabra, R. P., and Richardson, J. F., *Non-Newtonian Flow and Applied Rheology*, 2nd ed., Butterworth-Heinemann, Oxford, England, U.K., 2008.

[18] Bird, R. B., Armstrong, R. C., and Hassager, O., *Dynamics of Polymeric Liquids, Vol. 1: Fluid Dynamics*, 2nd ed., Wiley, New York, 1987.

[19] Laba, D., *Rheological Properties of Cosmetics and Toiletries*, Marcel-Dekker, New York, 1993.

[20] Steffe, J. F., *Rheological Methods in Food Process Engineering*, 2nd ed., Freeman Press, East Lansing, MI, 1996.

[21] Schramm, L., *Emulsions, Foams, Suspensions and Aerosols: Microscience and Applications*, 2nd ed., Wiley-VCH, New York, 2014.

[22] Paul, E. L., Atiemo-Obeng, V. A., and Kresta, S. M., *Handbook of Industrial Mixing: Science and Practice*, Wiley, New York, 2003.

[23] Rasschaert, F., Talansier, E., Blésès, D., Magnin, A., and Lambert, M., "Packaging of Yield Stress Fluids: Flow Patterns," *AIChE Journal*, Vol. 52, No. 5, 2018, pp. 1117–1126. doi:10.1002/aic.15989

[24] Amirian, E., Dejam, M., and Chen, Z., "Performance Forecasting for Polymer Flooding in Heavy Oil Reservoirs," *Fuel*, Vol. 216, No. 3, 2018, pp. 83–100. doi:10.1016/j.fuel.2017.11.110

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

Downloaded by COLORADO STATE UNIVERSITY on October 23, 2019 | http://arc.aiaa.org | DOI: 10.2514/1.T5614

[25] Saboorian-Jooybari, H., Dejam, M., and Chen, Z., "Heavy Oil Polymer Flooding from Laboratory Core Floods to Pilot Tests and Field Applications: Half-Century Studies," *Journal of Petroleum Science and Engineering*, Vol. 142, No. 6, 2016, pp. 85–100. doi:10.1016/j.petrol.2016.01.023

[26] Dejam, M., "Advective-Diffusive-Reactive Solute Transport due to Non-Newtonian Fluid Flows in a Fracture Surrounded by a Tight Porous Medium," *International Journal of Heat and Mass Transfer*, Vol. 128, No. 1, 2019, pp. 1307–1321. doi:10.1016/j.ijheatmasstransfer.2018.09.061

[27] Dejam, M., "Dispersion in Non-Newtonian Fluid Flows in a Conduit with Porous Walls," *Chemical Engineering Science*, Vol. 189, No. 11, 2018, pp. 296–310. doi:10.1016/j.ces.2018.05.058

[28] Prahashanna, A., and Chhabra, R. P., "Free Convection in Power-Law Fluids from a Heated Sphere," *Chemical Engineering Science*, Vol. 65, No. 23, 2010, pp. 6190–6205. doi:10.1016/j.ces.2010.09.003

[29] Gupta, A. K., Sasmal, C., Sairamu, M., and Chhabra, R. P., "Laminar and Steady Free Convection in Power-Law Fluids from a Heated Spheroidal Particle: A Numerical Study," *International Journal of Heat and Mass Transfer*, Vol. 75, No. 8, 2014, pp. 592–609. doi:10.1016/j.ijheatmasstransfer.2014.04.008

[30] Gupta, A. K., Gupta, S., and Chhabra, R. P., "Natural Convection in Bingham Plastic Fluids from an Isothermal Spheroid: Effects of Fluid Yield Stress, Viscous Dissipation and Temperature-Dependent Viscosity," *Korea-Australia Rheology Journal*, Vol. 29, No. 3, 2017, pp. 163–184. doi:10.1007/s13367-017-0018-y

[31] Nirmalkar, N., Gupta, A. K., and Chhabra, R. P., "Natural Convection from a Heated Sphere in Bingham Plastic Fluids," *Industrial and Engineering Chemistry Research*, Vol. 53, No. 45, 2014, pp. 17818–17832. doi:10.1021/ie503152k

[32] Nalluri, S. V., Patel, S. A., and Chhabra, R. P., "Mixed Convection from a Hemisphere in Bingham Plastic Fluids," *International Journal of Heat and Mass Transfer*, Vol. 84, No. 5, 2015, pp. 304–318. doi:10.1016/j.ijheatmasstransfer.2014.12.059

[33] Sasmal, C., and Chhabra, R.P., "Laminar Free Convection in Power-Law Fluids from a Heated Hemisphere," *Journal of Thermophysics and Heat Transfer*, Vol. 28, No. 4, 2014, pp. 750–763. doi:10.2514/1.T4394

[34] Hering, R. G., and Grosh, R. J., "Laminar Free Convection from a Non-Isothermal Cone," *International Journal of Heat and Mass Transfer*, Vol. 5, No. 11, 1962, pp. 1059–1068. doi:10.1016/0017-9310(62)90059-5

[35] Kuiken, H. K., "Axisymmetric Free Convection Boundary-Layer Flow Past Slender Bodies," *International Journal of Heat and Mass Transfer*, Vol. 11, No.7, 1968, pp. 1141–1153. doi:10.1016/0017-9310(68)90031-8

[36] Oosthuizen, P. H., and Wu, M. C. E., "A Numerical Study of Laminar Free Convective Heat Transfer from Vertical Cones," *Transactions of the Canadian Society for Mechanical Engineering*, Vol. 2, No. 1, 1973, pp. 9–14.

[37] Stewart, W. E., "Asymptotic Calculation of Free Convection in Laminar Three-Dimensional Systems," *International Journal of Heat and Mass Transfer*, Vol. 14, No. 8, Aug. 1971, pp. 1013–1031.

[38] Alamgir, M., "Over-all Heat Transfer from Vertical Cones in Laminar Free Convection: An Approximate Method," *Journal of Heat Transfer*, Vol. 101, No. 1, 1979, pp. 174–176. doi:10.1151/1.3450912

[39] Na, T. Y., and Chiou, J. P., "Laminar Natural Convection over a Slender Vertical Frustum of a Cone," *Wärme-und Stoffübertragung*, Vol. 12, No. 2, 1979, pp. 83–87. doi:10.1007/bf01002323

[40] Chiou, J. P., and Na, T. Y., "Laminar Natural Convection over a Slender Vertical Frustum of a Cone with Variable Surface Temperature," *Canadian Journal of Chemical Engineering*, Vol. 58, No. 4, 1980, pp. 438–442. doi:10.1002/cjce.5450580403

[41] Na, T. Y., and Chiou, J. P., "Laminar Natural Convection over a Frustum of a Cone," *Applied Scientific Research*, Vol. 35, No. 5, 1979, pp. 409–421. doi:10.1007/bf00420389

[42] Hasan, M., and Mujumdar, A. S., "Coupled Heat and Mass Transfer in Natural Convection Under Flux Condition Along a Vertical Cone," *International Communications in Heat and Mass Transfer*, Vol. 11, No. 2, 1984, pp. 157–172. doi:10.1016/0735-1933(84)90019-8

[43] Pop, I., and Watanabe, T., "Free Convection with Uniform Suction or Injection from a Vertical Cone for Constant Wall Heat Flux," *International Communications in Heat and Mass Transfer*, Vol. 19, No. 2, 1992, pp. 275–283. doi:10.1016/0735-1933(92)90038-j

[44] Pullepu, B., Ekambavanan, K., and Chamkha, A., "Unsteady Laminar Free Convection from a Vertical Cone with Uniform Surface Heat Flux," *Nonlinear Analysis: Modelling and Control*, Vol. 13, No. 1, 2008, pp. 47–60.

[45] Acrivos, A., "A Theoretical Analysis of Laminar Natural Convection Heat Transfer to Non-Newtonian Fluids," *AIChE Journal*, Vol. 6, No. 4, 1960, pp. 584–590. doi:10.1002/aic.690060416

[46] Shenoy, A., "Laminar Natural Convection Heat Transfer from a Slender Vertical Cone to a Power–Law Fluid," *Canadian Journal of Chemical Engineering*, Vol. 61, No. 6, 1983, pp. 869–872. doi:10.1002/cjce.5450610615

[47] Marner, W. J., and Kleppe, J., "Combined Free and Forced Laminar Convection in a Bingham Plastic on a Vertical Flat Plate," *Journal of Fluids Engineering*, Vol. 98, No. 1, 1976, pp. 119–121. doi:10.1115/1.3448164

[48] Kleppe, J., and Marner, W. J., "Transient Free Convection in a Bingham Plastic on a Vertical Flat Plate," *Journal of Heat Transfer*, Vol. 94, No. 4, 1972, pp. 371–376. doi:10.1115/1.3449954

[49] Shenoy, A., *Heat Transfer to Non-Newtonian Fluids: Fundamentals and Analytical Expressions*, Wiley-VCH, New York, 2018.

[50] Oosthuizen, P. H., and Donaldson, E., "Free Convective Heat Transfer from Vertical Cones," *Journal of Heat Transfer*, Vol. 94, No. 3, 1972, pp. 330–331. doi:10.1115/1.3449945

[51] Ashjaee, M., Arzaghi, M., Jarrahi, M., and Yousefi, T., "Experimental and Numerical Study of Free Convection on an Isothermal Downward Cone," *Experimental Heat Transfer*, Vol. 20, No. 4, 2007, pp. 307–322. doi:10.1080/08916150701418286

[52] Mishra, P., Gupta, S., and Chhabra, R. P., "Effect of Orientation on Drag Characteristics of a Cone Settling in Power-Law Fluids," *Journal of Chemical Engineering of Japan*, Vol. 52, No. 1, 2019, pp. 19–30. doi:10.1252/jcej.17we297

[53] Papanastasiou, T. C., "Flows of Materials with Yield," *Journal of Rheology*, Vol. 31, No. 5, 1987, pp. 385–404. doi:10.1122/1.549926

[54] Mitsoulis, E., and Tsamopoulos, J., "Numerical Simulations of Complex Yield-Stress Fluid Flows," *Rheologica Acta*, Vol. 56, No. 3, 2017, pp. 231–258. doi:10.1007/s00397-016-0981-0

[55] O' Donovan, E. J., and Tanner, R. I., "Numerical Study of the Bingham Squeeze Film Problem," *Journal of Non-Newtonian Fluid Mechanics*, Vol. 15, No. 1, 1984, pp. 75–83. doi:10.1016/0377-0257(84)80029-4

[56] Bercovier, M., and Engelman, M., "A Finite-Element Method for Incompressible Non-Newtonian Flows," *Journal of Computational Physics*, Vol. 36, No. 3, 1980, pp. 313–326. doi:10.1016/0021-9991(80)90163-1

[57] Baskharone, E. A., *The Finite Element Method with Heat Transfer and Fluid Mechanics Applications*, Cambridge Univ. Press, New York, 2013.

[58] Macosko, C.W., *Rheology: Principles, Measurements and Applications*, Wiley, New York, 1994.

[59] Amato, W. S., and Tien, C., "Free Convection Heat Transfer from Isothermal Spheres in Polymer Solutions," *International Journal of Heat and Mass Transfer*, Vol. 19, No. 11, 1976, pp. 1257–1266. doi:10.1016/0017-9310(76)90077-6

[60] Thompson, R. L., and Soares, E. J., "Viscoplastic Dimensionless Numbers," *Journal of Non-Newtonian Fluid Mechanics*, Vol. 238, No. 12, 2016, pp. 57–64. doi:10.1016/j.jnnfm.2016.05.001

[61] Li, C., Magnin, A., and Metivier, C., "Natural Convection in Shear-Thinning Yield Stress Fluids in a Square Enclosure," *AIChE Journal*, Vol. 62, No. 4, 2016, pp. 1347–1355. doi:10.1002/aic.15112